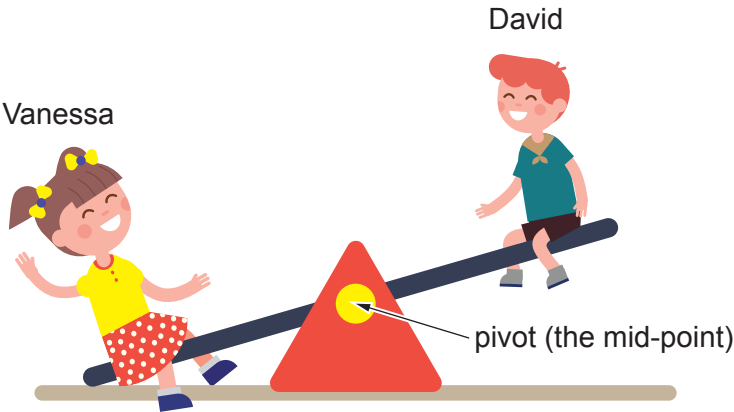


Answer **all** questions.

1. (a) The picture shows Vanessa and David sitting on a see-saw.



The see-saw in the diagram is not balanced because Vanessa is heavier than David.

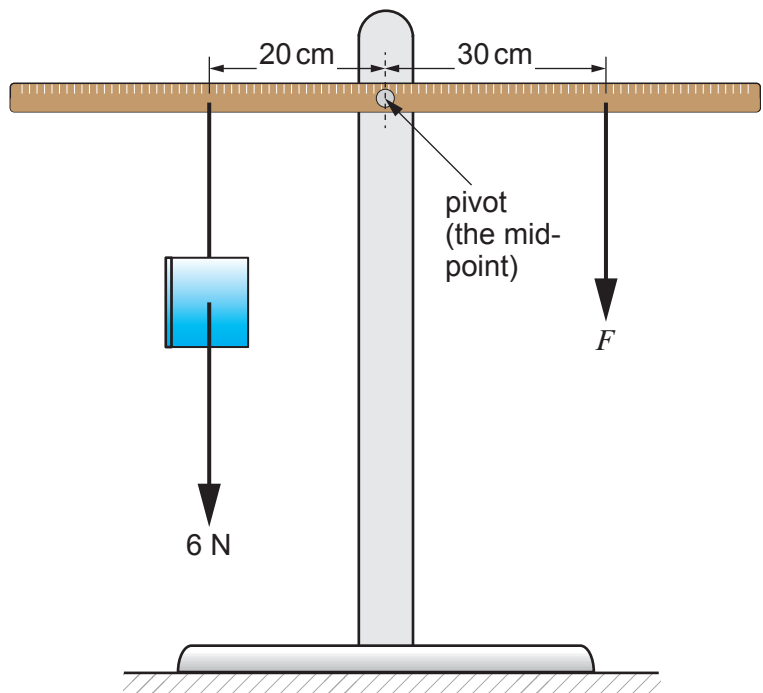
Place a **tick (✓)** in **one** box alongside the correct statement below. [1]

- Vanessa and David can't balance the see-saw because they have different weights. ☐
- If Vanessa moves nearer the pivot, the see-saw can be balanced. ☐
- For the see-saw to balance, they **must** sit at the ends. ☐



Examiner  
only

- (b) A metre ruler is hung from its mid-point. A book weighing 6 N is 20 cm from the pivot. It is balanced by a force,  $F$  as shown in the diagram.



Use the equation:

$$F = \frac{6 \times 20}{\text{distance}}$$

to calculate the value of the force,  $F$ .

[2]

force,  $F =$  ..... N

|   |
|---|
|   |
| 3 |

